

Reachability Distance and Local Outlier Factor

- For two objects, o and o' , the reachability distance from o' to o is $dist(o \leftarrow o')$ if $dist(o, o') > dist_k(o)$; $dist_k(o)$ otherwise, that is,
$$reachdist_k(o \leftarrow o') = \max\{dist_k(o), dist(o, o')\}$$
- Reachability distance is not symmetric, that is, in general, $reachdist_k(o \leftarrow o') \neq reachdist_k(o' \leftarrow o)$
- The local reachability density of an object, o , is

$$lrd_k(o) = \frac{|N_k(o)|}{\sum_{o' \in N_k(o)} reachdist_k(o' \leftarrow o)}$$

- The local outlier factor of an object o is

$$LOF_k(o) = \frac{\sum_{o' \in N_k(o)} \frac{lrd_k(o')}{lrd_k(o)}}{|N_k(o)|} = \sum_{o' \in N_k(o)} lrd_k(o') \cdot \sum_{o' \in N_k(o)} reachdist_k(o' \leftarrow o)$$